The Journal of Open Source Software

MultiVae: A Python package for Multimodal Variational Autoencoders on Partial Datasets.

Agathe Senellart [1,2,3¶], **Clément Chadebec**[1,2,3], **and Stéphanie Allassonnière**[1,2,3]

1 Université de Paris-Cité **2** Inria **3** Inserm ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Øystein Sørensen

Reviewers:

- @rudraprsd
- @maruti-iitm
- @powerfulbean

Submitted: 14 March 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

In recent years, multimodal machine learning has seen significant growth, especially in representation learning and data generation. Recently, Multimodal Variational Autoencoders (VAEs) have been attracting growing interest for both tasks, thanks to their versatility, scalability, and interpretability as latent variable models. They are particularly useful in *partially observed* settings, such as medical applications, where available datasets are often incomplete (Antelmi et al., 2019; Lawry Aguila et al., 2023).

We introduce MultiVae, an open-source Python library offering unified implementations of multimodal VAEs. It is designed for easy and customizable use of these models on fully or partially observed data. It facilitates the development and benchmarking of new algorithms by including standard benchmark datasets, evaluation metrics and tools for monitoring and sharing models.

Multimodal Variational Autoencoders

Multimodal VAEs aim to: (1) Learn a shared representation from multiple modalities; (2) Generate one missing modality from available ones.

These models learn a latent representation z of all modalities in a lower dimensional space and learn to *decode* z to generate each modality. Let $X = (x_1, x_2, ...x_M)$ contain M modalities. In the VAE setting, we define an *encoder* distribution $q_\phi(z|X)$ projecting the observations to the latent space, and decoders distributions $(p_\theta(x_i|z))_{1\leq i\leq M}$ translating the latent code z back to observations. Those distributions are parameterized by neural networks that are trained to minimize an objective function derived from variational inference. See Kingma & Welling (2014) to learn more about the VAE framework and Suzuki & Matsuo (2022) for a survey on multimodal VAEs.

A key differentiator of multimodal VAEs relies in the choice of the encoder $q_\phi(z|X)$. They fall into three main categories: *Aggregated models* (Shi et al., 2019; Sutter et al., 2021; Wu & Goodman, 2018) use a mean or product operation to combine modalities, *Joint models* (Senellart et al., 2023; Suzuki et al., 2016; Vedantam et al., 2018) use a neural network taking all modalities as input, and *Coordinated models* (Tian & Engel, 2019; Wang et al., 2017) use separate latent spaces with additional similarity constraints.

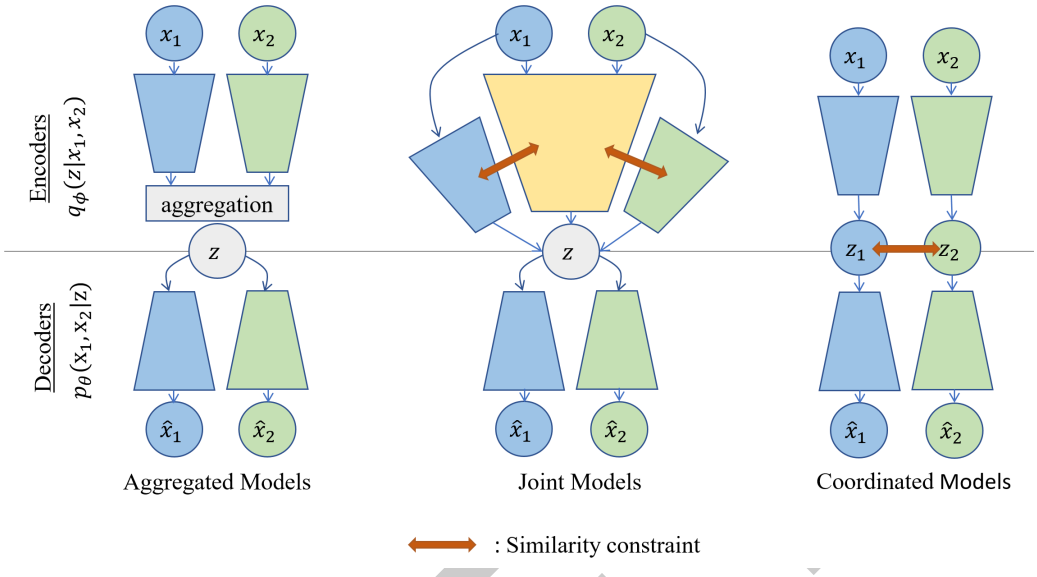

Figure 1: Different types of multimodal VAEs

MultiVae unifies these approaches in a modular and extensible way. Notably, aggregated models offer a natural way of *learning* on incomplete datasets: for an incomplete sample X, the encoding z and the objective function can be computed using only available modalities. MultiVae is the first library to provide implementations of these models with built-in support for missing data, using masks during loss computation.

Data Augmentation

Another application of VAEs is Data Augmentation (DA): by sampling new latent codes z and decoding them, *fully synthetic multimodal* samples can be generated to augment a dataset. This approach has been successfully used with unimodal VAEs to augment datasets for data-intensive deep learning applications (Chadebec et al., 2023). However, it remains underexplored in the multimodal setting. MultiVae includes a `multivae.samplers` module with several sampling strategies to further explore the generative abilities of these models.

Statement of Need

Despite the usefulness of multimodal VAEs, the lack of easy-to-use and verified implementations might hinder applicative research. MultiVae offers unified implementations, designed to be accessible even for non-specialists. We ensure reliability by reproducing key results from original papers whenever possible.

Related libraries contain implementations of Multimodal VAEs: the Multimodal VAE Comparison Toolkit (Sejnova et al., 2024), Pixyz (Masahiro Suzuki & Matsuo, 2023) and multi-view-ae (Aguila et al., 2023) that is most closely related to us and released while we were developing MultiVae.

We compare in a summarizing table below, the different features of each work. MultiVae differs and complements existing software packages in key ways: it supports **incomplete datasets**, which we consider essential for real-life applications, as well as **generative samplers**, **benchmark datasets** and **metrics** to facilitate research. It contains a large range of models with a great flexibility on parameters' choices and including all implementation details present in the original codes that improve performance.

DRAFT

List of Models and Features

We list models and features in each work. Symbol (✓*) indicates that the implementation includes additional options unavailable in the others.

Models/ Features	Ours	Aguila et al. (2023)	Sejnova et al. (2024)	Masahiro Suzuki & Matsuo (2023)
JMVAE (Suzuki et al., 2016)	✓*	✓		✓
MVAE (Wu & Goodman, 2018)	✓*	✓	✓	✓
MMVAE (Shi et al., 2019)	✓*	✓	✓	
MoPoE (Sutter et al., 2021)	✓*	✓	✓	
DMVAE (Lee & Pavlovic, 2021)	✓	✓*	✓	
MVTCAE (Hwang et al., 2021)	✓	✓		
MMVAE+ (Palumbo et al., 2023)	✓*	✓		
CMVAE (Palumbo et al., 2024)	✓			
Nexus (Vasco et al., 2022)	✓			
CVAE (Kingma & Welling, 2014)	✓			✓
MHVAE (Dorent et al., 2023)	✓			
TELBO (Vedantam et al., 2018)	✓			
JNF (Senellart et al., 2023)	✓			
CRMVAE (Suzuki & Matsuo, 2023)	✓			
MCVAE (Antelmi et al., 2019)		✓		
mAAE		✓		
DVCCA (Wang et al., 2017)		✓		
DCCAE (Wang et al., 2015)		✓		
mWAE		✓		
mmJSD (Sutter et al., 2020)		✓		
gPoE (Lawry Aguila et al., 2023)		✓		
Support of Incomplete datasets	✓			
GMM Sampler	✓			
MAF Sampler, IAF Sampler	✓			
Metrics: {Likelihood, Coherences, FIDs, Reconstruction, Clustering}	✓			
Benchmark Datasets	✓		✓	
Model sharing via Hugging Face	✓			

Code Quality and Documentation

MultiVae is available on GitHub and Pypi, with full documentation at https://multivae.readthedocs.io/. The code is unit-tested with 94% coverage. We provide **tutorials** either as notebooks or scripts allowing users to get started easily. To further showcase how to use our library for research applications, we provide detailed *case studies* in the documentation.

Acknowledgements

We are grateful to the authors of the initial implementations of the models included in MultiVae. This work benefited from state grant managed by the Agence Nationale de la Recherche under the France 2030 program, AN-23-IACL-0008. This research has been partly supported by the European Union under the (2023-2030) ERC Synergy Grant 101071601.

DRAFT

References

Aguila, A. L., Jayme, A., Montaña-Brown, N., Heuveline, V., & Altmann, A. (2023). Multi-view-AE: A python package for multi-view autoencoder models. *Journal of Open Source Software*, *8*(85), 5093. https://doi.org/10.21105/joss.05093

Antelmi, L., Ayache, N., Robert, P., & Lorenzi, M. (2019). *Sparse multi-channel variational autoencoder for the joint analysis of heterogeneous data*. *97*, 302–311. https://proceedings.mlr.press/v97/antelmi19a.html

Chadebec, C., Thibeau-Sutre, E., Burgos, N., & Allassonnière, S. (2023). Data augmentation in high dimensional low sample size setting using a geometry-based variational autoencoder. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(3), 2879–2896. https://doi.org/10.1109/TPAMI.2022.3185773

Dorent, R., Haouchine, N., Kogl, F., Joutard, S., Juvekar, P., Torio, E., Golby, A. J., Ourselin, S., Frisken, S., Vercauteren, T., Kapur, T., & Wells, W. M. (2023). Unified brain MR-ultrasound synthesis using multi-modal hierarchical representations. In *Medical image computing and computer assisted intervention – MICCAI 2023* (pp. 448–458). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-43999-5_43

Hwang, H., Kim, G.-H., Hong, S., & Kim, K.-E. (2021). Multi-view representation learning via total correlation objective. *Advances in Neural Information Processing Systems*, *34*, 12194–12207.

Kingma, D. P., & Welling, M. (2014). *Auto-Encoding Variational Bayes*. arXiv. http://arxiv.org/abs/1312.6114

Lawry Aguila, A., Chapman, J., & Altmann, A. (2023). *Multi-modal variational autoencoders for normative modelling across multiple imaging modalities*. 425–434. https://doi.org/10.1007/978-3-031-43907-0_41

Lee, M., & Pavlovic, V. (2021). *Private-shared disentangled multimodal VAE for learning of latent representations*. 1692–1700. https://doi.org/10.1109/CVPRW53098.2021.00185

Masahiro Suzuki, T. K., & Matsuo, Y. (2023). Pixyz: A python library for developing deep generative models. In *Advanced Robotics* (No. 0; Vol. 0, pp. 1–16). Taylor & Francis. https://doi.org/10.1080/01691864.2023.2244568

Palumbo, E., Daunhawer, I., & Vogt, J. E. (2023). MMVAE+: Enhancing the generative quality of multimodal VAEs without compromises. *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=sdQGxouELX

Palumbo, E., Manduchi, L., Laguna, S., Chopard, D., & Vogt, J. E. (2024). *Deep generative clustering with multimodal diffusion variational autoencoders*. https://openreview.net/forum?id=k5THrhXDV3

Sejnova, G., Vavrecka, M., Stepanova, K., & Taniguchi, T. (2024). *Benchmarking multimodal variational autoencoders: CdSprites+ dataset and toolkit*. https://arxiv.org/abs/2209.03048

Senellart, A., Chadebec, C., & Allassonnière, S. (2023). Improving multimodal joint variational autoencoders through normalizing flows and correlation analysis. *arXiv Preprint arXiv:2305.11832*.

Shi, Y., Siddharth, N., Paige, B., & Torr, P. H. S. (2019). Variational Mixture-of-Experts Autoencoders for Multi-Modal Deep Generative Models. *arXiv:1911.03393 [Cs, Stat]*. http://arxiv.org/abs/1911.03393

Sutter, T. M., Daunhawer, I., & Vogt, J. E. (2020). Multimodal generative learning utilizing jensen-shannon-divergence. *CoRR*, *abs/2006.08242*. https://arxiv.org/abs/2006.08242

DRAFT

Sutter, T. M., Daunhawer, I., & Vogt, J. E. (2021). Generalized Multimodal ELBO. *ICLR*.

Suzuki, M., & Matsuo, Y. (2022). A survey of multimodal deep generative models. *Advanced Robotics*, *36*(5-6), 261–278. https://doi.org/10.1080/01691864.2022.2035253

Suzuki, M., & Matsuo, Y. (2023). *Mitigating the limitations of multimodal VAEs with coordination-based approach*. https://openreview.net/forum?id=Rn8u4MYgeNJ

Suzuki, M., Nakayama, K., & Matsuo, Y. (2016). Joint Multimodal Learning with Deep Generative Models. *arXiv:1611.01891 [Cs, Stat]*. http://arxiv.org/abs/1611.01891

Tian, Y., & Engel, J. (2019). Latent Translation: Crossing Modalities by Bridging Generative Models. *ArXiv*.

Vasco, M., Yin, H., Melo, F. S., & Paiva, A. (2022). Leveraging hierarchy in multimodal generative models for effective cross-modality inference. *Neural Networks*, *146*, 238–255.

Vedantam, R., Fischer, I., Huang, J., & Murphy, K. (2018). Generative Models of Visually Grounded Imagination. *arXiv:1705.10762 [Cs, Stat]*. http://arxiv.org/abs/1705.10762

Wang, W., Arora, R., Livescu, K., & Bilmes, J. (2015). On deep multi-view representation learning. In F. Bach & D. Blei (Eds.), *Proceedings of the 32nd international conference on machine learning* (Vol. 37, pp. 1083–1092). PMLR. https://proceedings.mlr.press/v37/wangb15.html

Wang, W., Yan, X., Lee, H., & Livescu, K. (2017). *Deep Variational Canonical Correlation Analysis*. https://doi.org/10.48550/arXiv.1610.03454

Wu, M., & Goodman, N. (2018). Multimodal Generative Models for Scalable Weakly-Supervised Learning. *Advances in Neural Information Processing Systems*, *31*. https://proceedings.neurips.cc/paper/2018/hash/1102a326d5f7c9e04fc3c89d0ede88c9-Abstract.html

DRAFT